Final Project: Building a General-Purpose Reasoning Agent

Requirements

- Build a general-purpose reasoning agent
  - Solve arbitrary problem-solving tasks
  - Use inference-time algorithms to implement an agent

- Implement at least 8 distinct inference-time algorithms or techniques
  - CoT, Self-Consistency, Tree of Thought, Self-Refine, ReACT, Decomposition, Tool-augmented Reasoning etc.
  - Any other techniques not discussed in class

Core Constraints

- Use only the provided LLM API
  - No external LLMs (e.g., GPT, Claude, etc.)
  - No paid APIs or search services
  - Agent must run without additional costs

- Do not hardcode full delegation to external tools

- Do not use AI coding assistants (Cursor etc.) to implement the project

Efficiency Requirements

- ≤ 20 LLM calls per question
  - Final test set contains 1000+ instances
  - Up to 10% penalty of the project grade for inefficient LLM usage

- Limited LLM call budget during development
- Use calls wisely and optimize early

Evaluation Criteria

- Correctness on final test set
  - Efficiency (LLM call limits respected)
  - Quality and diversity of techniques
  - Code clarity and reproducibility

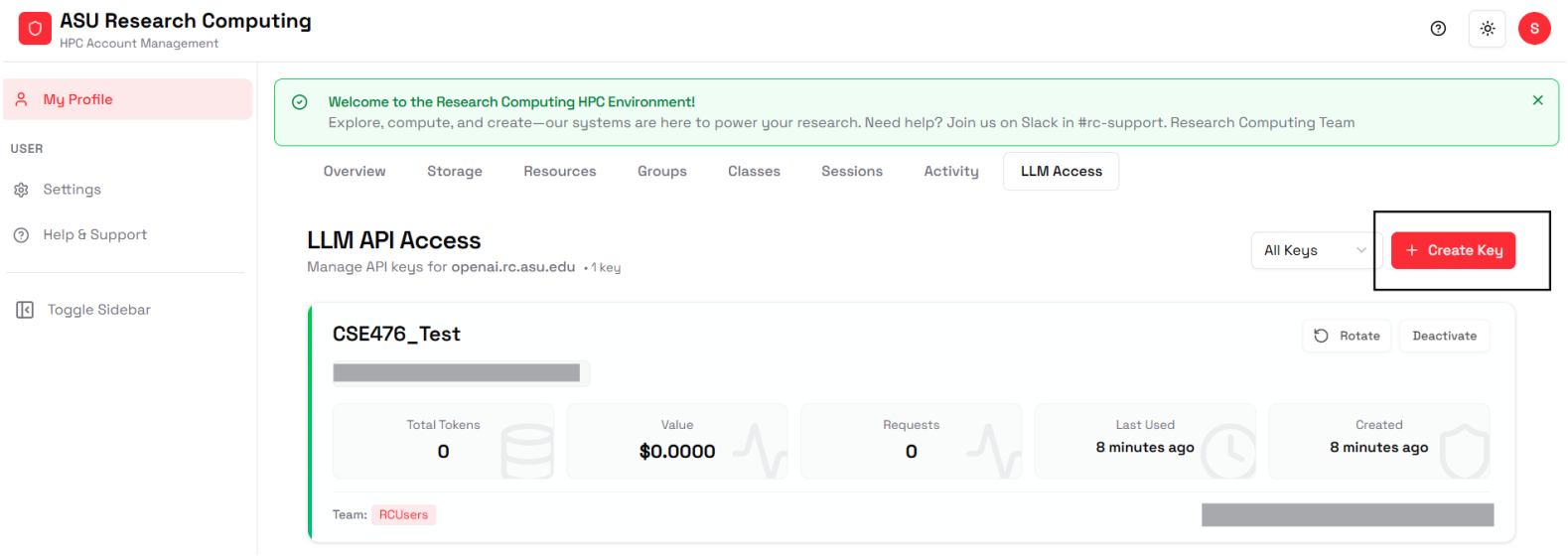
ASU Research Compute (SOL)

- You need to be in university network to use Research Compute
  - Install Cisco VPN
    - Accessing SOL
- Request has been raised for everyone in the class for SOL access.
- Reference Only (You may not need these)
  - SOL Wiki
  - Working with Python

ASU Research Compute (SOL)

- You will need these
  - Create an API Key. Instruction
  - We will use “qwen3-30b-a3b-instruct-2507” model to evaluate your code. For final submission verify that your code works with this model
  - You are free to explore other models and their capabilities

ASU Research Compute (SOL)



ASU Research Compute (SOL)

Model Directory

Optimized overview of every deployed model without the clutter.

8 / 8 visible

Search by model or provider

MODE All modes

llama4-scout-17b Gaudi2	General	Context 66K	Usage (30d) —	Calls (30d) —	
qwen3-coder-30b-a3b-instruct Gaudi2	General	Context 131K	Usage (30d) —	Calls (30d) —	
qwen3-30b-a3b-thinking-2507 Gaudi2	General	Context 131K	Usage (30d) —	Calls (30d) —	
qwen3-30b-a3b-instruct-2507 Gaudi2	General	Context 131K	Usage (30d) 182	Calls (30d) 3	
qwen3-235b-a22b-instruct-2507 Gaudi2	General	Context 262K	Usage (30d) —	Calls (30d) —	
qwen3-235b-a22b-thinking-2507 Gaudi2	General	Context 262K	Usage (30d) —	Calls (30d) —	
qwen3-embedding-8b Gaudi2	General	Context 33K	Usage (30d) —	Calls (30d) —	
qwen25-vl-7b Gaudi2	General	Context 33K	Usage (30d) —	Calls (30d) —	

ASU Research Compute (SOL)

Directory • Usage • Benchmarks

Explore the directory catalog, inspect usage logs, and compare LiveBench benchmarks in one place.

Directory Catalog | Usage Logs | Benchmarks

Period: Last 30 day

Date	Key	Model	Tokens	Value
Mar 17, 5:49 PM	CSE476_Test	qwen3-30b-a3b-instruct-2507	48	$0.0000
Mar 17, 5:48 PM	CSE476_Test	qwen3-30b-a3b-instruct-2507	48	$0.0000
Mar 17, 5:39 PM	CSE476_Test	qwen3-30b-a3b-instruct-2507	86	$0.0001
Mar 17, 5:38 PM	CSE476_Test	qwen3-30b-a3b-instruct-250	0	$0.00